

MAXIM

10 μ A, Low-Dropout, Precision Voltage References

MAX872/MAX874

General Description

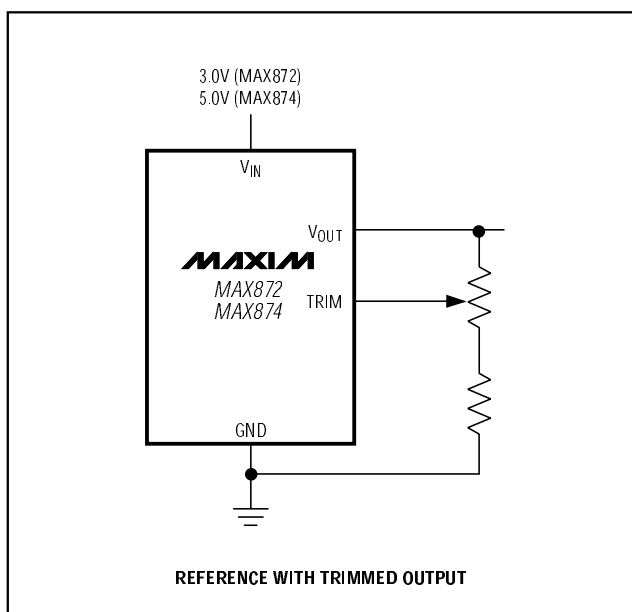
The MAX872/MAX874 precision 2.5V and 4.096V micro-power voltage references consume a maximum of only 10 μ A and operate from supply voltages up to 20V. The combination of ultra-low quiescent current and low 200mV dropout makes them ideal for battery-powered equipment. They source and sink up to 500 μ A with only 200mV input voltage headroom, which makes the 2.5V MAX872 ideal for use with a 3V supply and the 4.096V MAX874 ideal for use with a 5V supply.

Initial accuracy of 0.2% at +25°C (± 5 mV for the MAX872, ± 8 mV for the MAX874) and low 40ppm/°C max drift make these references suitable for a wide range of precision applications.

Applications

Hand-Held Instruments
Battery-Operated Equipment
Power Supplies

Typical Operating Circuit



Features

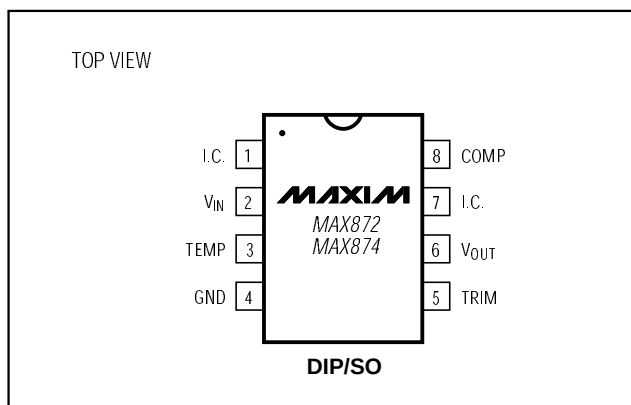
- ◆ **Output Voltage**
2.500V $\pm 0.2\%$ (MAX872)
4.096V $\pm 0.2\%$ (MAX874)
- ◆ **Wide Operating Voltage Range**
2.7V to 20V (MAX872)
4.3V to 20V (MAX874)
- ◆ **10 μ A Max Supply Current**
- ◆ **40ppm/°C Max Drift Over Extended Temp. Range**
- ◆ **Line Regulation Over Temp.**
20 μ V/V (MAX872)
75 μ V/V (MAX874)
- ◆ **Load Regulation Over Temp.**
0.6mV/mA Max (MAX872)
1.0mV/mA Max (MAX874)
- ◆ **$\pm 500\mu$ A Sink/Source Current**

Ordering Information

PART	TEMP. RANGE	PIN-PACKAGE
MAX872CPA	0°C to +70°C	8 Plastic DIP
MAX872CSA	0°C to +70°C	8 SO
MAX872C/D	0°C to +70°C	Dice*
MAX872EPA	-40°C to +85°C	8 Plastic DIP
MAX872ESA	-40°C to +85°C	8 SO
MAX874CPA	0°C to +70°C	8 Plastic DIP
MAX874CSA	0°C to +70°C	8 SO
MAX874C/D	0°C to +70°C	Dice*
MAX874EPA	-40°C to +85°C	8 Plastic DIP
MAX874ESA	-40°C to +85°C	8 SO

* Dice are specified at +25°C only.

Pin Configuration



MAXIM

Maxim Integrated Products 1

For free samples & the latest literature: <http://www.maxim-ic.com>, or phone 1-800-998-8800.
For small orders, phone 1-800-835-8769.

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ABSOLUTE MAXIMUM RATINGS

Supply Voltage 24V
 Output Short-Circuit Duration Continuous to Either Supply
 CCOMP Input -0.3V to V_{OUT}
 TRIM Input -0.3V to ($V_{IN} + 0.3V$)
 TEMP Output -0.3V to ($V_{IN} + 0.3V$)
 Continuous Power Dissipation ($T_A = +70^\circ\text{C}$)
 Plastic DIP (derate 9.09mW/ $^\circ\text{C}$ above $+70^\circ\text{C}$) 727mW
 SO (derate 5.88mW/ $^\circ\text{C}$ above $+70^\circ\text{C}$) 471mW

Operating Temperature Ranges

MAX87_C__ 0°C to $+70^\circ\text{C}$
 MAX87_E__ -40°C to $+85^\circ\text{C}$
 Storage Temperature Range -65°C to $+150^\circ\text{C}$
 Junction Temperature Range (T_j) -65°C to $+160^\circ\text{C}$
 Lead Temperature (soldering, 10sec) $+300^\circ\text{C}$

Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

ELECTRICAL CHARACTERISTICS—MAX872

($V_{IN} = 2.7V$, $I_L = 0\text{mA}$, $T_A = +25^\circ\text{C}$, unless otherwise noted.)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Output Voltage	V_{OUT}		2.495	2.500	2.505	V
Output Voltage Noise	e_n	0.1Hz to 10Hz		60		$\mu\text{Vp-p}$
Line Regulation	V_{OUT}/V_{IN}	$V_{IN} = 4.5V$ to $20V$		4	12	$\mu\text{V/V}$
		$V_{IN} = 2.7V$ to $5.5V$		80	250	
Load Regulation (Note 1)	V_{OUT}/I_{OUT}	Sourcing 0mA to 0.5mA		0.2	0.5	mV/mA
		Sinking 0mA to -0.5mA		4	12	
Quiescent Supply Current	I_Q			6.5	10	μA
Change in Supply Current vs. V_{IN}	I_Q/V_{IN}	$V_{IN} = 2.7V$ to $20V$		0.35	0.55	$\mu\text{A/V}$
Short-Circuit Output Current	I_{SC}	V_{OUT} short to GND		6	15	mA
		V_{OUT} short to V_{IN}		3	9	
TEMP Voltage	V_{TEMP}			690		mV
V_{OUT} Adjustment Range	V_{ADJ}	$V_{OUT} \geq V_{OUT} + 0.2V$	+75/-20	+100/-25		mV

ELECTRICAL CHARACTERISTICS—MAX874

($V_{IN} = 4.3V$, $I_L = 0\text{mA}$, $T_A = +25^\circ\text{C}$, unless otherwise noted.)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Output Voltage	V_{OUT}		4.088	4.096	4.104	V
Output Voltage Noise	e_n	0.1Hz to 10Hz		90		$\mu\text{Vp-p}$
Line Regulation	V_{OUT}/V_{IN}	$V_{IN} = 4.3V$ to $20V$		15	75	$\mu\text{V/V}$
Load Regulation (Note 1)	V_{OUT}/I_{OUT}	Sourcing 0mA to 0.5mA		0.15	0.9	mV/mA
		Sinking 0mA to -0.5mA		6	15	
Quiescent Supply Current	I_Q			6.5	10	μA
Change in Supply Current vs. V_{IN}	I_Q/V_{IN}	$V_{IN} = 4.3V$ to $20V$		0.35	0.55	$\mu\text{A/V}$
Short-Circuit Output Current	I_{SC}	V_{OUT} short to GND		6	15	mA
		V_{OUT} short to V_{IN}		5	25	
TEMP Voltage	V_{TEMP}			690		mV
V_{OUT} Adjustment Range	V_{ADJ}	$V_{OUT} \geq V_{OUT} + 0.2V$	± 150	± 200		mV